

Post-Entry Markouts of Position-Increasing Hyperliquid Taker Decisions

Aggregate markout structure and a cross-wallet persistence experiment

Quantitative Research — Internal

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1. Executive summary

This report measures post-entry markouts of position-increasing taker decisions on Hyperliquid (BTC, ETH, SOL, HYPE) and asks whether persistent cross-wallet differences in markout identify informed or copyable traders. The metric is a post-fill decision markout, $\text{direction} \times (\text{close}(\text{entry} + h) / \text{close}(\text{entry}) - 1)$ in basis points, priced at the first bar strictly after the timestamp. It is a signed price move after a decision — not realized PnL, and, with candle prices and no order-book data, not a full execution-quality markout. Only position-increasing, taker-crossed, non-wash fills are scored; closing fills are excluded from scoring; same-bar same-direction opening fills are aggregated; sub-hour execution markouts are unavailable and the shortest horizon is one hour.

Four distinct objects are reported and should not be conflated:

1. **Unconditional aggregate markout** is near zero through eight hours and highly dispersed; per-coin intervals mostly include zero and directional hit rates are near 50%.
2. **Dynamic monthly wallet ranking** shows modest next-month persistence: next-month markout rises across prior-month rank deciles, and top-decile membership recurs more often than a permutation null over the training window.
3. A **training-defined fixed recurring cohort** (121 wallets) outperforms the non-cohort field under two of three weighting schemes, but the advantage is not resolved under equal-wallet weighting.
4. **Benchmark-adjusted wallet value** is not established: the cohort's return is close to a mechanical reversal benchmark evaluated at the same timestamps, and the incremental fade-adjusted difference is not statistically resolved under any weighting.

Entries are, on average, contrarian — placed after moves that ran against the eventual trade direction — and subsequent markout declines with the signed pre-entry move (pooled slope -0.058 bp per bp, 95% CI $[-0.108, -0.018]$). The overall picture is consistent with recurring exposure to short-horizon reversal conditions; a smaller wallet-specific effect remains possible but is unresolved in this sample. Because the validation window informed later specification choices, Part II is exploratory; a prospective test with execution measurement is required.

Part I — Aggregate taker markouts

2. Data and methodology

Universe. 7,332,013 taker entries from 32,949 wallets, 2025-08-01 to 2026-06-30 (UTC), restricted to BTC, ETH, SOL, HYPE. A mid-frequency cohort is frozen on identity, not performance ($\geq \$200$ majors fills, taker share ≥ 0.70 , median round-trip hold 1–24 h, $\geq \$200$ training entries): 1,694 wallets, 812,616 priced entries.

Observation unit. One (wallet, coin, bar, direction) position-increasing taker decision. Maker, wash and closing fills are tracked for position accounting but not scored. Same-bar same-direction opening fills aggregate to one entry; scaling across bars yields multiple entries.

Markout and adjustment. Horizons 1, 2, 4, 8, 24 h, priced post-fill. Raw markout is the price move; drift-neutralized markout subtracts, per coin-month and horizon, that coin's own mean forward return. Neutralization is the adjusted measure reported; a beta/market-adjusted measure is defined but deferred.

Cost schedule (canonical). All cost-adjusted figures use one table — **BTC 8, ETH 8, SOL 10, HYPE 14 bp round-trip** — applied per scored entry (one round-trip charge per decision, conservative for a one-way markout). The schedule is a modelling assumption, not a bound; it excludes latency slippage, size-dependent impact, funding, partial execution and overlapping-capital constraints. Results are reported as “cost-adjusted under the assumed schedule,” never as “net.”

Weighting. Term-structure and distribution results are entry-weighted; persistence and decile tests wallet-day-weighted; the fixed cohort is reported under three weightings (§9).

Inference. Confidence intervals use a **calendar-day cluster bootstrap** (unique days drawn independently with replacement, 2,000 draws, wallets and coins carried with their day). Calendar-day clustering preserves contemporaneous dependence across wallets and coins; it does **not** preserve serial dependence across adjacent days. For 24-hour outcomes a 5-day non-overlapping moving-block variant addresses serial dependence and is reported alongside. Multiplicity uses Benjamini–Hochberg FDR.

Effective sample. The 812,616 entries span 334 calendar days, 1,694 wallets, 179,402 wallet-days and 252,223 wallet-coin-days. Because the resampling unit is the calendar day ($\approx \$334$ clusters), intervals are wide despite the large entry count.

Splits. Training < 2026-02-01; embargo February 2026; validation 2026-03-01 to 06-30. The validation window was examined during the study, so Part II validation results are exploratory.

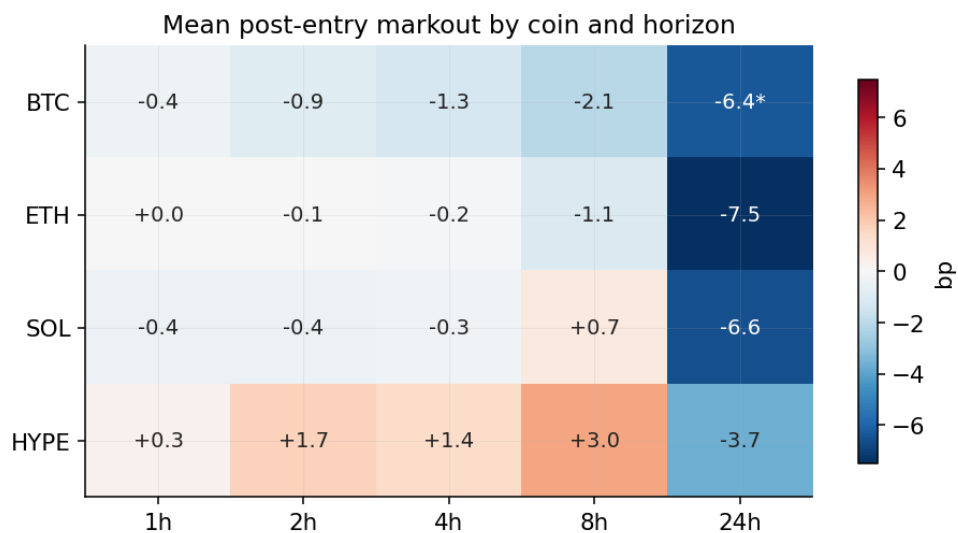
3. Aggregate markout term structure

Mean markout is small and its interval is wide relative to the mean at every horizon through 8 hours; hit rates are near 50%.

Coin	Horizon	Mean bp	95% CI	Median	Hit rate	Std bp	N
				bp			
BTC	1h	−0.37	[−1.1, +0.3]	+0.0	49.9%	65	356,077

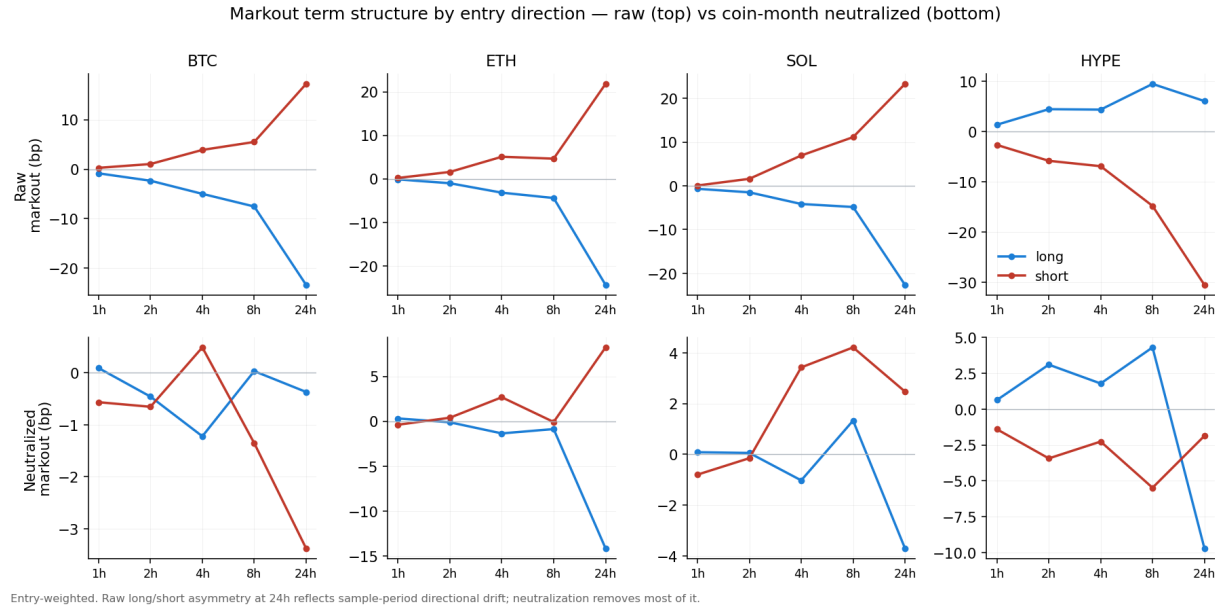
Coin	Horizon	Mean bp	95% CI	Median bp	Hit rate	Std bp	N
BTC	8h	-2.08	[-5.7, +1.1]	+0.1	50.0%	151	355,894
BTC	24h	-6.37	[-13.5, 0.0]†	-3.2	49.3%	250	355,371
ETH	8h	-1.09	[-7.7, +5.2]	+0.9	50.2%	226	195,786
SOL	8h	+0.74	[-7.1, +8.4]	+3.5	50.7%	258	135,299
HYPE	8h	+2.96	[-6.6, +12.9]	-0.3	49.9%	332	125,207
ALL	8h	-0.60	[-5.4, +3.9]	+0.7	50.2%	224	812,186
ALL	24h	-6.26	[-16.5, +3.4]	-5.7	49.1%	363	811,321

†BTC 24h is the only horizon whose interval approaches zero; the 5-day moving-block interval is [-13.0, +0.1], so it is marginal rather than resolved. (Full 4×4 table in the appendix.)



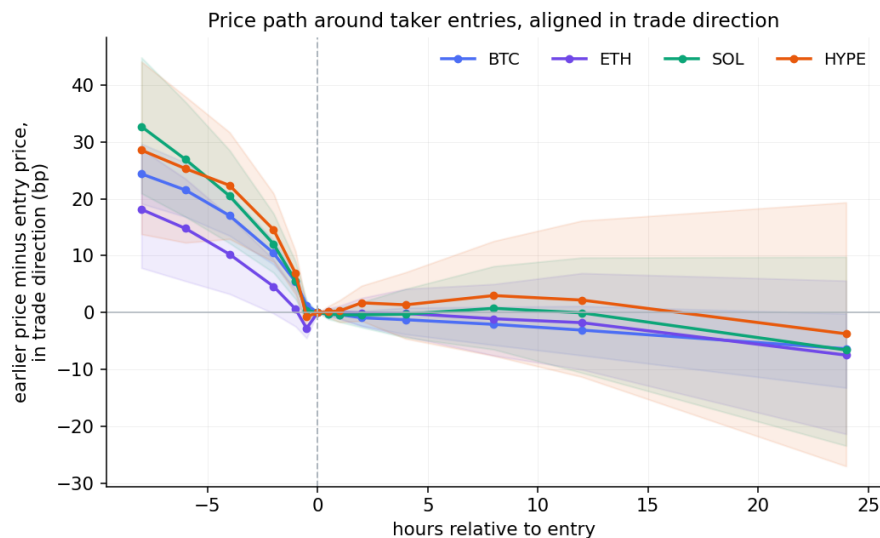
Full sample, entry-weighted, bp. * = 95% calendar-day cluster interval excludes zero.

Raw long and short markouts are strongly asymmetric at 24 hours (BTC long -23.4 vs short +17.2 bp), consistent with sample-period directional drift; after coin-month neutralization the asymmetry is materially smaller (BTC -0.4 vs -3.4, largest residual in ETH). The 24-hour figures are largely coin trend rather than entry timing.



4. Conditional markouts

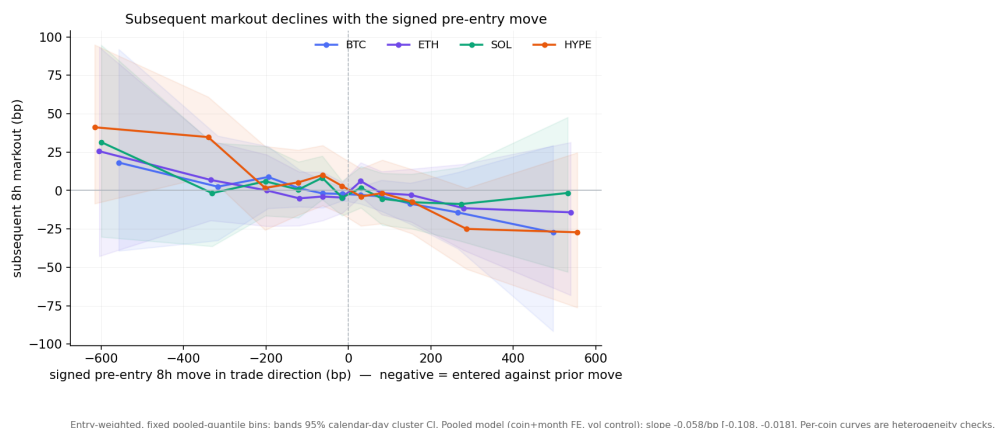
Aligning entries at their decision bar in trade-direction terms, the pre-entry price is above the entry price for longs and below it for shorts (about +18 to +33 bp at -8 h) and returns to the entry price by construction at $t = 0$. The event study establishes only that **prices tended to move against the eventual trade direction before entry**; because the curve is normalized to the entry price it cannot, on its own, establish exhaustion.



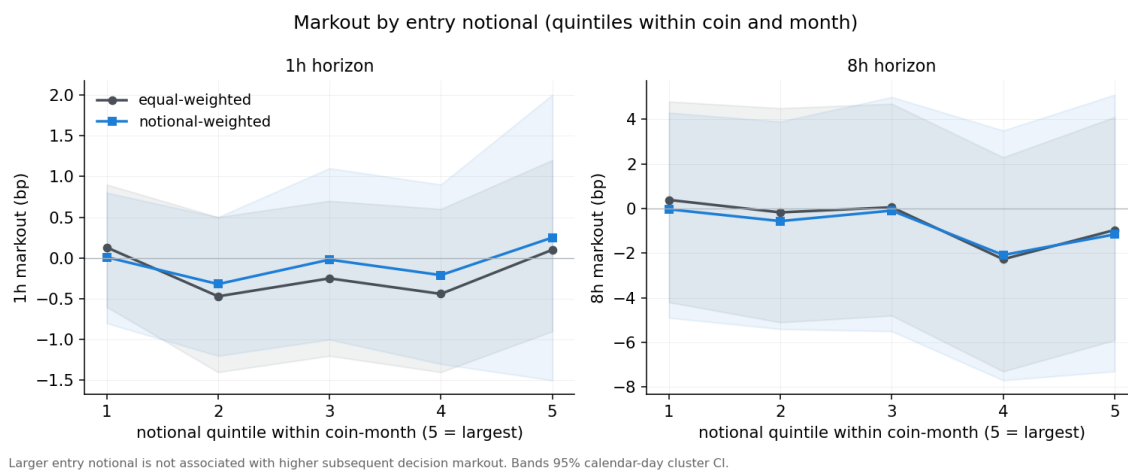
All cohort entries, entry-weighted. Left of 0 = before entry. Positive pre-entry = price was above (long) / below (short) the entry price, i.e. it moved against the eventual trade direction into entry. Right of 0 = subsequent markout. 95% calendar-day cluster CI.

Reading the pre-entry axis. The value at offset t is $\text{dir} \times (\text{close}(\text{entry}+t)/\text{close}(\text{entry}) - 1)$. For a long, a positive value at $t = -8$ h means the earlier price was above the entry price (price fell into entry); for a short, that the earlier price was below it (price rose into entry). Positive pre-entry values thus indicate the price moved against the eventual trade direction.

Conditioning subsequent 8-hour markout on the signed pre-entry move, a pooled model (markout $\sim$ signed pre-entry move + coin fixed effects + month fixed effects + realized volatility, day-cluster bootstrap, $n = 590,910$) gives a slope of **-0.058 bp per bp, 95% CI [-0.108, -0.018]** (two-sided interval excludes zero). Per-coin slopes are all negative and are shown as heterogeneity checks; their one-sided tests of $H_0: \beta \geq 0$ are each below 0.05, but with four coin tests and the widest uncertainty on BTC we do not claim each coin is individually established. Subsequent markout generally declines as the pre-entry move turns from against to with the trade — consistent with short-horizon reversal exposure.

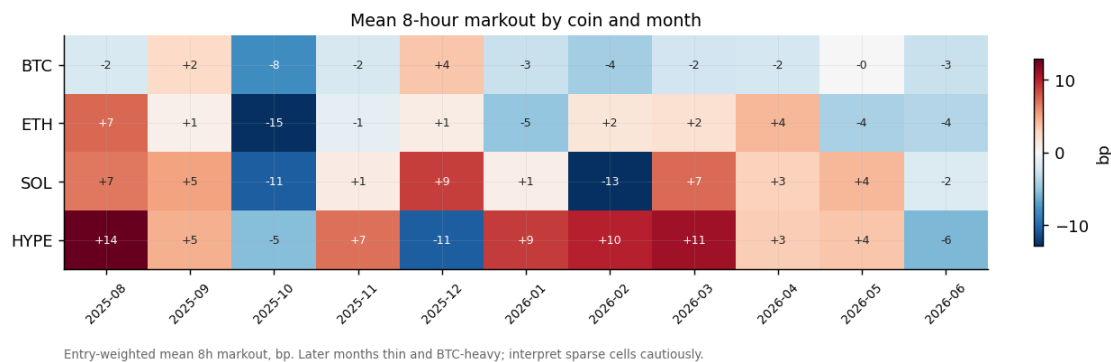


Forming notional quintiles within coin and month, larger entries do not show higher markout; equal-weighted and notional-weighted means are similar and the largest quintile is not the highest. Larger entry notional is not associated with higher subsequent decision markout in this specification. (Candle data cannot measure price impact; no impact claim is made.)



5. Stability, distributions and execution limitations

Mean 8-hour markout varies month to month and the panel thins and becomes BTC-heavy in later months.



The per-entry distribution is wide and near-symmetric: at 8 hours the interquartile range is roughly ± 75 to ± 190 bp by coin, so the small means sit well inside the dispersion (appendix). Markout by realized-volatility quintile and time of day is in the appendix.

Execution-analysis limitation. This report uses candle prices. A full execution-markout analysis requires actual fill price, contemporaneous midprice, spread, order-book depth and immediate price impact. With order-book data a future study would add fill-to-mid implementation shortfall, immediate markout at the seconds-to-minutes scale, spread paid, price impact versus trade size, and adverse selection versus depth. None is measured here; the markout reflects post-fill candle-to-candle movement only, and cost-adjusted figures assume the fixed schedule above.

Part II — Search for informed wallets

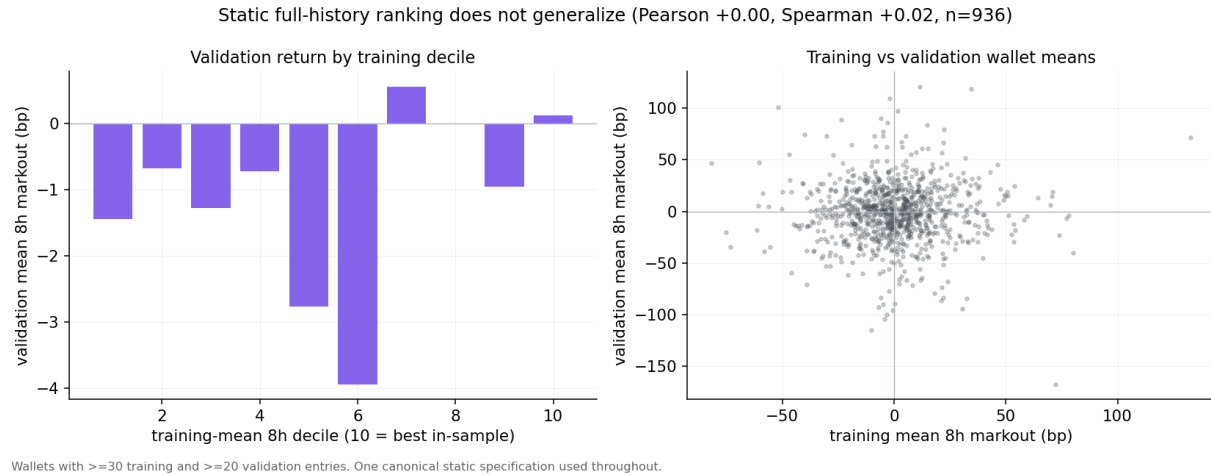
Aggregate markout is small relative to per-entry dispersion, but pooled estimates can conceal informed subgroups. We test whether cross-wallet differences are persistent and useful for selection or copy trading. Three wallet objects are kept explicitly separate: a **static** full-history ranking, a **dynamic** monthly ranking, and a **fixed** recurring cohort.

6. Cross-wallet heterogeneity

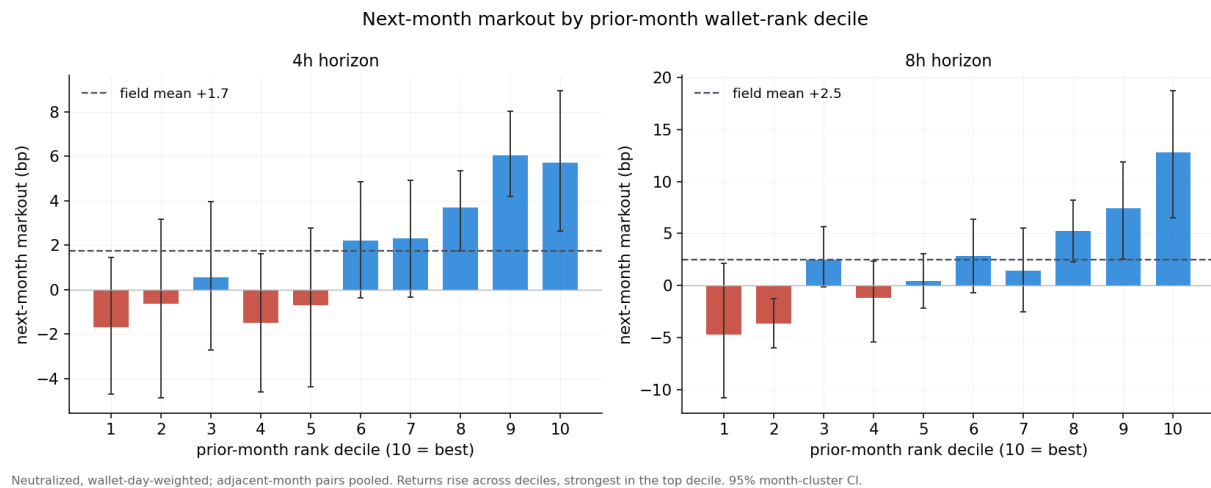
Wallet-level mean markout is dispersed, but much of the spread is sampling noise: high in-sample means concentrate among wallets with fewer entries, and selecting wallets on best in-sample markout returns -28.7 bp in validation. Top in-sample wallets (appendix) reach $+40$ to $+130$ bp gross at 8 h — descriptive in-sample values, not evidence of a repeatable edge.

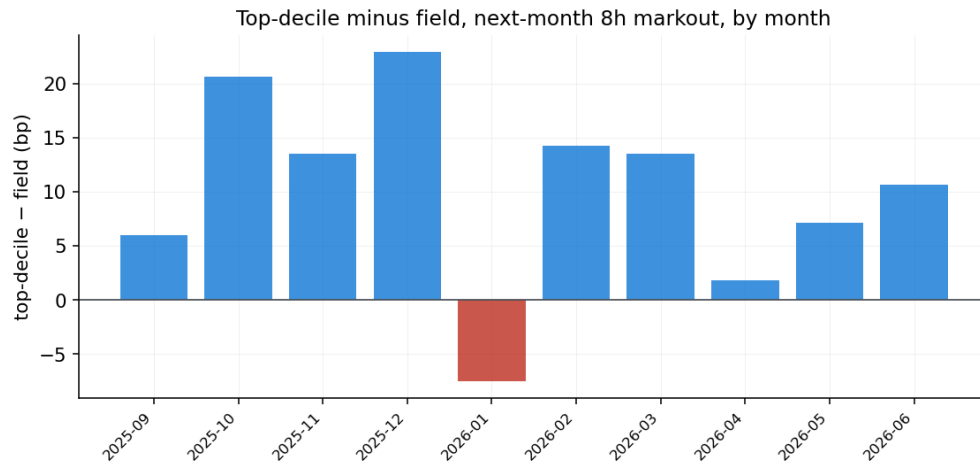
7. Static and rolling persistence

Under **static** ranking (one canonical specification: wallets with $\geq \$30$ training and $\geq \$20$ validation entries, raw 8-hour wallet means, $n = 936$), the training-to-validation correlation is Pearson $+0.00$ / Spearman $+0.02$ — no generalization. The highest training deciles do not retain an advantage, and a walk-forward selection across nine metrics returned a best coin p-value of 0.09 with no FDR survivors.



Under **dynamic** monthly ranking, next-month markout rises across prior-month rank deciles — a gradient, strongest in the top decile, not solely a top-decile effect — and the prior-month top decile leads the field by about +4.0 bp at 4 h and +10.3 bp at 8 h. The adjacent-month rank correlation of neutralized, wallet-day-weighted markout is +0.093 at 8 h ([+0.061, +0.126], positive in 10 of 10 folds).





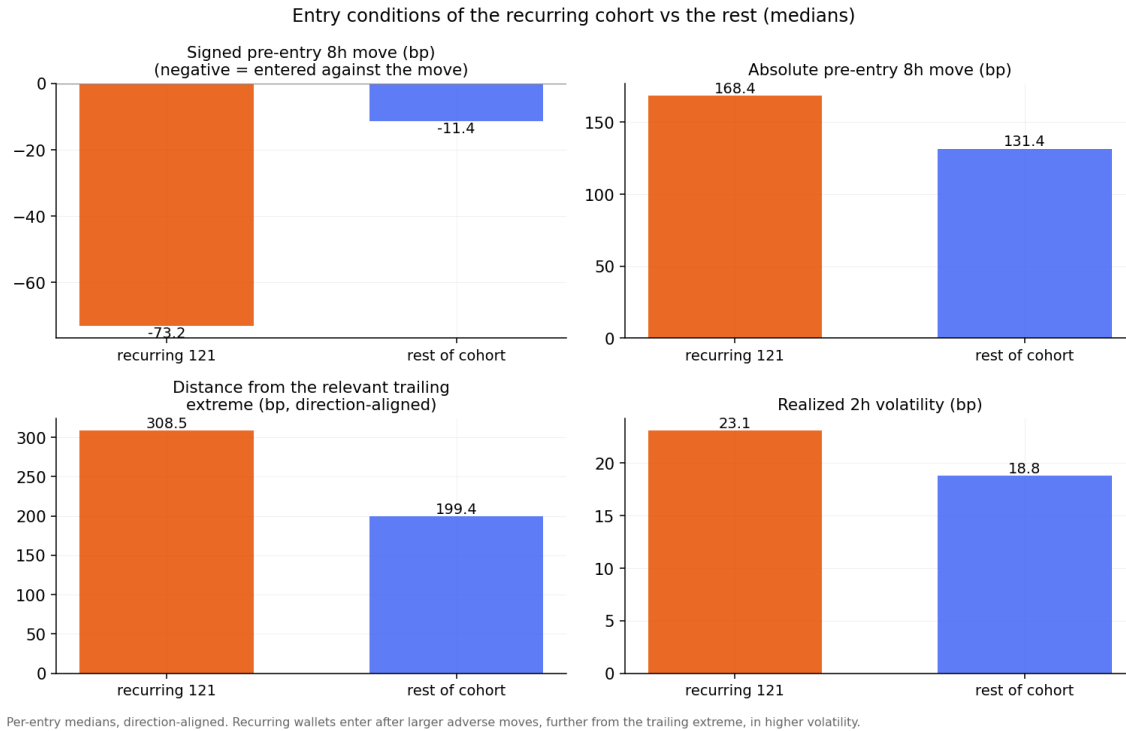
Neutralized wallet-day-weighted. Positive in 9 of 10 months.

Recurrence. Over the six training months (the only window that may justify the cohort construction), top-decile membership recurs more than a permutation null preserving each month's eligible set and slot count allows: $\geq \$2$ months 121 vs 88.7, $\geq \$3$ months 27 vs 8.3, $\geq \$4$ months 5 vs 0.5 (all $p < 0.001$). The same test over all 11 months (213/72/25 vs 185/35/5) agrees but overlaps the validation window and is reported as exploratory only.

8. Recurring-wallet characterization

We freeze a recurring cohort as wallets in the top decile in at least two of the six training months (121 wallets), on drift-neutralized 8-hour markout with wallet-day weighting. Recurrence itself is statistically supported (§7). These wallets are distinguished by conviction and cross-coin breadth, not size or activity (mean training rank percentile 0.72 vs 0.48; four coins vs three; near-identical median ticket).

Their entries occur after larger adverse moves, further from the relevant trailing extreme, and in higher volatility than the rest of the cohort (direction-aligned medians below).



At the wallet level this contrarian tendency is common but not universal: 83% of the 121 have a net-contrarian entry record and 65% fade the trailing move on at least 60% of entries (appendix).

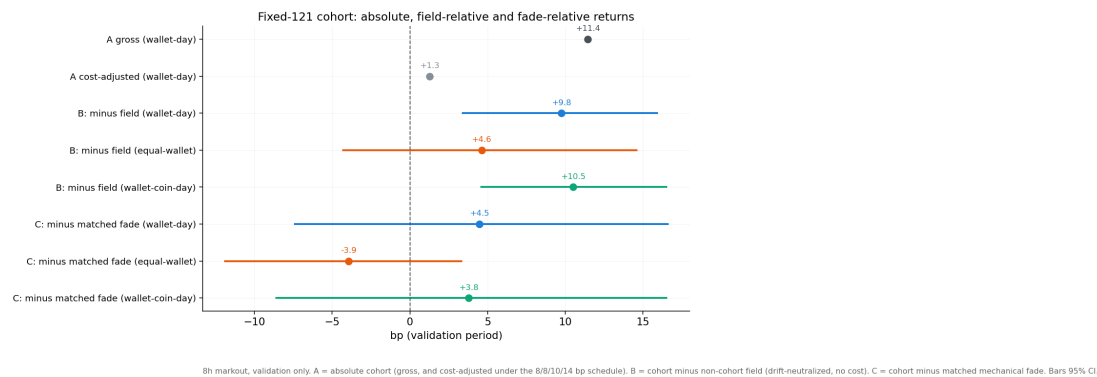
9. Benchmark, copyability and the fixed cohort

We compare high-markout wallets to a mechanical reversal benchmark (fade = $-\text{sign}(\text{trailing } 8\text{-hour move})$) at the same timestamps, so execution effects cancel in the difference. For the training-ranked top decile at 8 h, gross return is +14.0 bp ([+5.4, +22.8]) and cost-adjusted +7.1 bp; the fade-adjusted difference is +1.0 bp ([−10.8, +13.7], $p = 0.44$). Trading in the wallet's direction did not improve on the fade. Conditioning the fade return on observable market state leaves a per-event residual of −1.3 bp ([−15.6, +13.0]) and a wallet-day-capped residual of +5.9 bp ([−5.5, +17.7]); the minimum effect this validation sample can resolve is about 11–16 bp, so a smaller incremental effect cannot be excluded but none is established. Adding wallet identity, rank, direction and size to a market-state model changed cross-fit predictive correlation by −0.0003.

Fixed 121-cohort — three estimands, three weightings. The cohort has 121 wallets; 97 were active in validation (24 attrition — no scored entries in the window), spanning 4,286 wallet-days and 5,811 wallet-coin-days over 122 calendar days. We report the absolute cohort return (A), the difference versus the non-cohort field (B, drift-neutralized, no cost — a relative measure, not a long-short strategy), and the difference versus the matched fade (C).

Metric	Wallet-day (primary)	Equal-wallet	Wallet-coin-day (sens.)
Active wallets	97	97	97
A. Absolute gross (bp)	+11.4	+4.4	+12.0
A. Absolute cost-adjusted (bp)	+1.3	−5.7	+2.0

Metric	Wallet-day (primary)	Equal-wallet	Wallet-coin-day (sens.)
B. Minus non-cohort field (bp)	+9.8	+4.6	+10.5
B. 95% CI	[+3.4, +15.9]	[−4.3, +14.6]	[+4.6, +16.5]
B. p-value	0.001	0.169	0.001
C. Minus matched fade (bp)	+4.5	−3.9	+3.8
C. p-value	0.25	0.16	0.26



The field-relative advantage (B) is resolved under the two day-based weightings (+9.8 / +10.5, $p = 0.001$) but falls to +4.6 ($p = 0.169$, interval includes zero) under equal-wallet weighting, which does not reward the cohort's defining cross-coin breadth; the absolute cost-adjusted return is small-positive under the day schemes and negative under equal-wallet. The incremental value over the matched fade (C) is not statistically resolved under any weighting ($p = 0.16$ – 0.26). Concentration diagnostics: the field advantage concentrates in SOL (+12.5 bp) versus BTC (+2.2); one of four validation months is negative (−8.6 bp in May); excluding the single largest-contributing wallet leaves $B = +9.7$ bp (unchanged). Diagnostics only; the cohort is not re-selected on validation performance.

A prospective test is required to separate reversal exposure from wallet skill: on new data, under identical capital constraints and a fixed 8-hour horizon, compare a market-state reversal model, the same model augmented with recurring-wallet activity and direction, and direct copying of wallet direction — logging execution, spread, fees and funding.

10. Conclusion

Aggregate post-entry markout is small relative to per-entry dispersion through eight hours, with a marginal negative 24-hour drift concentrated in BTC. A negative relationship is present between the signed pre-entry move and subsequent markout, consistent with short-horizon reversal exposure.

Dynamic wallet rankings and the fixed recurring cohort exhibit positive field-relative markout in historical validation. However, the estimated return is similar to a mechanical reversal benchmark evaluated at the same timestamps, and the available sample does not statistically resolve incremental wallet-direction value. The result is therefore consistent with recurring exposure to reversal conditions, while a smaller wallet-specific effect remains possible. The field-relative advantage is also weighting-dependent, and the fixed cohort remains exploratory because its validation window in-

formed specification choices. Prospective data with actual execution measurement is required to evaluate copyability.

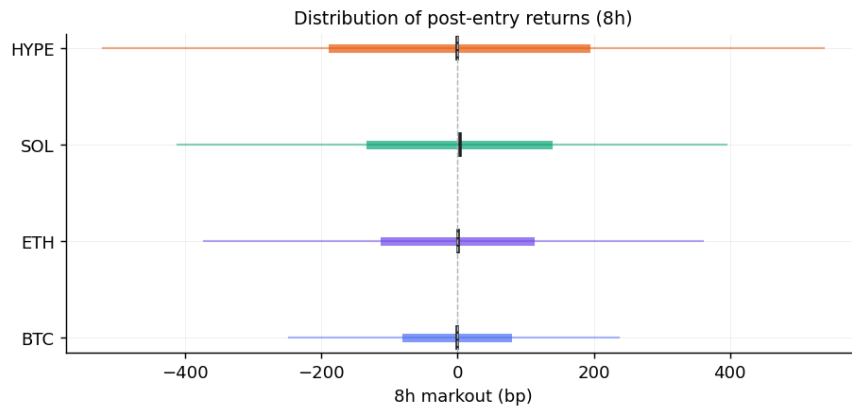
Appendix

A. Full markout term-structure table (calendar-day cluster 95% CI)

Coin	Horizon	Mean bp	95% CI	Median bp	Hit rate	Std bp	N
BTC	1h	-0.37	[-1.1, +0.3]	+0.0	49.9%	65	356,077
BTC	4h	-1.27	[-3.7, +0.8]	+0.2	50.1%	116	356,030
BTC	8h	-2.08	[-5.7, +1.1]	+0.1	50.0%	151	355,894
BTC	24h	-6.37	[-13.5, 0.0]	-3.2	49.3%	250	355,371
ETH	1h	+0.00	[-1.1, +1.2]	+0.0	49.9%	95	195,851
ETH	4h	-0.15	[-4.3, +4.1]	+0.3	50.1%	174	195,839
ETH	8h	-1.09	[-7.7, +5.2]	+0.9	50.2%	226	195,786
ETH	24h	-7.49	[-20.6, +5.4]	-8.8	48.7%	363	195,672
SOL	1h	-0.42	[-1.8, +1.0]	+0.0	49.9%	109	135,350
SOL	4h	-0.28	[-4.7, +4.3]	+1.5	50.4%	194	135,344
SOL	8h	+0.74	[-7.1, +8.4]	+3.5	50.7%	258	135,299
SOL	24h	-6.58	[-24.6, +9.4]	-5.4	49.4%	411	135,223
HYPE	1h	+0.29	[-1.5, +2.1]	+0.5	50.1%	131	125,267
HYPE	4h	+1.35	[-4.6, +7.2]	-1.2	49.7%	240	125,259
HYPE	8h	+2.96	[-6.6, +12.9]	-0.3	49.9%	332	125,207
HYPE	24h	-3.68	[-27.3, +21.4]	-16.9	48.6%	537	125,055
ALL	8h	-0.60	[-5.4, +3.9]	+0.7	50.2%	224	812,186
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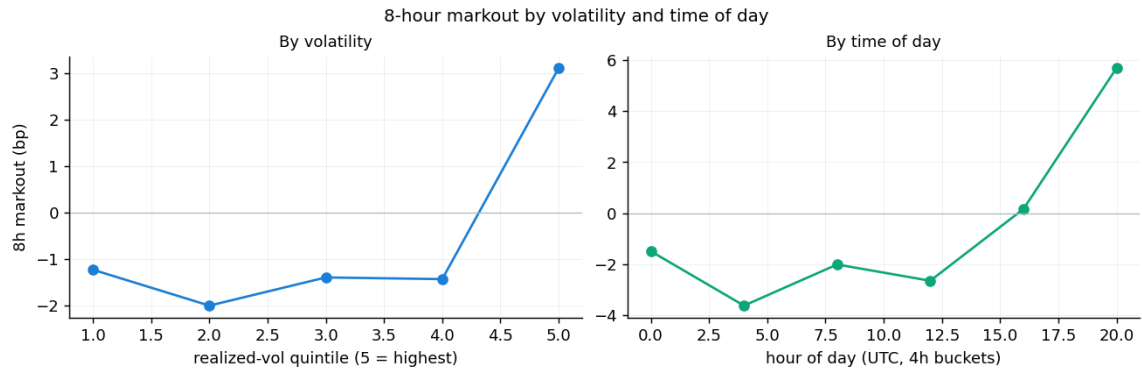
B. Trade-size and dispersion quantiles

Trade notional (\$): BTC median 4,386 (p90 94,608); ETH 4,306 (101,699); SOL 2,906 (45,930); HYPE 2,500 (42,710). 8-hour markout dispersion (bp, p10 / median / p90 / std): BTC -172 / +0 / +167 / 151; ETH -258 / +1 / +249 / 226; SOL -286 / +4 / +283 / 258; HYPE -381 / -0 / +398 / 332.

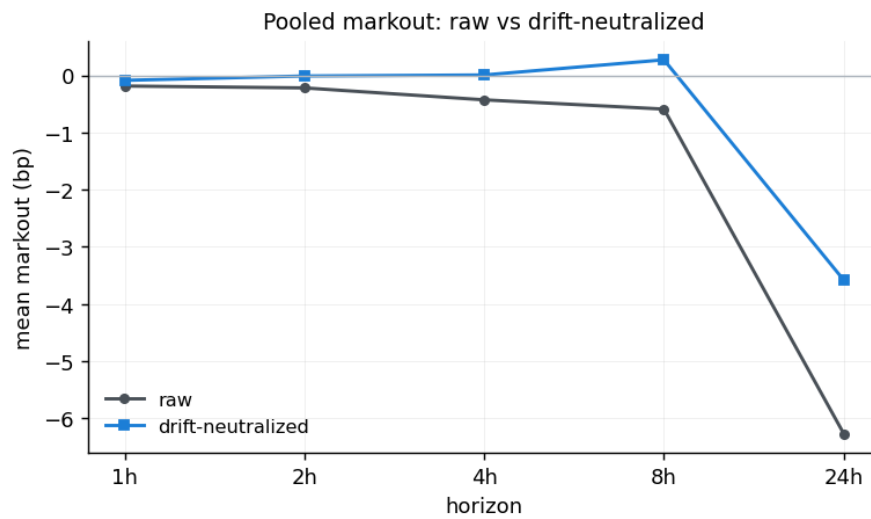


Thin line: 5th-95th percentile; thick bar: IQR; marker: median. Entry-weighted.

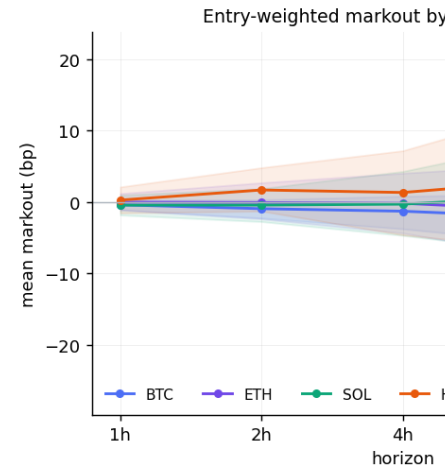
C. Secondary conditioning and raw-vs-neutralized



Entry-weighted, pooled over coins.

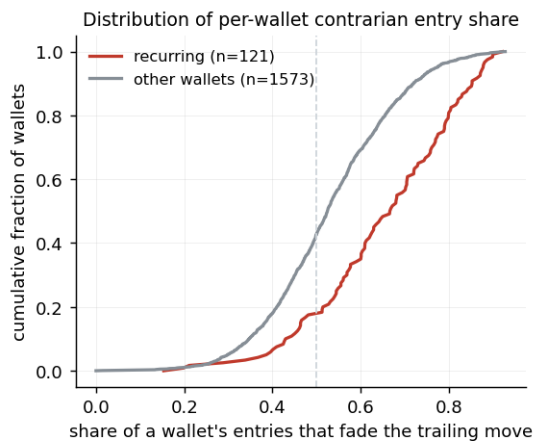


Pooled over four coins, entry-weighted. Neutralized = raw minus each coin's own monthly mean forward return.



Full sample, entry-weighted. Bands: 95% calendar-day

D. Wallet-level detail



Per wallet, ≥ 20 training entries. Contrarian = entry direction opposite the trailing 8h move. Dashed line = 50% (coin-flip).

Top training-decile wallets (in-sample descriptive; do not persist per §7):

#	wallet	N	mean 8h (bp)	Sharpe	hit%	med \$
1	0x5aadb434...	214	+132.8	0.36	69	12,061
2	0xfd490cf8...	236	+124.2	0.65	76	131,053
3	0xdfb34089...	248	+83.3	0.33	64	5,535
4	0x172f5a11...	202	+80.2	0.16	51	3,225
5	0x1ecc7c22...	200	+78.9	0.29	61	25,028

Per-trade Sharpe (mean/std of markout, un-annualized).

A companion technical appendix documents the audit trail, the number-to-source mapping and the reproducibility package.